

II BE EXAMINATION DEC 2023
Computer Engineering
3ACRC1 Applied Mathematics-III

Time-3 hours

Maximum Marks [60]

Note-Attempt all the questions with any two of (a), (b) and (c) from each.

- Q.1 (a) Perform seven iterations to find the smallest positive real root, correct to four places of decimal, of the equation using Regula-falsi method- [6]

$$\sin x - e^x \cos x = 0$$

- (b) Perform seven iterations of Gauss-Seidel method to find solution of the system- [6]

$$3.2x - 6.1y + 2.4z = 1.5$$

$$-7.3x - 3y + 1.8z = -2.8$$

$$2.5x + 1.7y - 7.4z = -3.9$$

- (c) Find $y(1.20)$ in two steps from $\frac{dy}{dx} = \frac{x(2 \ln x + 1)}{\sin y + y \cos y}$, $y(0.50) = 0.85$, using Modified Euler's method. [6]

- Q.2 (a) The latent heat r of vaporization of steam is given in the following table at different temperatures t :

t	40	50	60	70	80
r	1069.1	1063.6	1058.2	1052.7	1049.3

Find the rate of change of latent heat at $t = 76$. [6]

- (b) Compute the value of the integral $\int_{\pi/2}^{\pi} e^{2x} \cos \ln x dx$ using Simpson's 1/3 and 3/8 rules taking 13 ordinates. [6]

- (c) The following table gives the viscosity V of an oil as a function of temperature $t(^{\circ}C)$. Find the viscosity V of oil at the temperature $135^{\circ}C$ using Newton's divided difference formula. [6]

t	110	130	140	180	200
V	10.8	8.1	5.5	4.8	4.2

- Q.3 (a) A household appliance manufacturer wants to analyze the relationship between total sales and the company's two primary means of advertising television and magazines from the following data-

X	67	72	63	65	71	64
Y	22	25	29	31	29	32
Z	33	40	25	30	35	29

where X = Total sales, Y = expenditure on television, Z = expenditure on radio. All amounts are in lakhs. Calculate the total sales corresponding to the expenditures $X_2 = 34$, $X_3 = 20$. [6]

- (b) The proportion of the budget for a certain type of industrial company that is allotted to environmental and pollution control is coming under scrutiny. A data collection project determines that the distribution of these proportions is given by-

$$f(x) = \begin{cases} 4(1+x), & 0 \leq x < 1/4 \\ \frac{2}{9}(1-4x^2), & 1/4 \leq x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Find the distribution function and using it determine the probability that a company chosen at random expends more than 20% but less than 35% its budget on environmental and pollution controls? [6]

- (c) In an engineering examination, a student is considered to have failed, secured second class, first class and distinction, according as the score is less than 45%, between 45% and 60%, between 60% and 75% and above 75% respectively. In a particular year, 10% of the students failed in the examination and 5% got distinction. Assuming the normal distribution of the marks, find the percentages of the students who got first class and second class. [6]

- Q.4 (a) In a test of the quality of two television commercials, each commercial was shown in a separate test area six times over a one-week period. The following week a telephonic survey was conducted to identify individuals who had seen the commercials. Those individuals were asked to state the primary message in the commercials. The following results were recorded:

	Commercial A	Commercial B
Number who saw commercial	150	200
Number who recalled message	63	60

Use $\alpha = 0.05$ and test the hypothesis that there is no difference in the recall proportions for the two commercials. [$|z_{0.05}| = 1.96$] [6]

- (b) The following data shows the cost in hundreds rupees per square meter of the floor area concerning randomly selected 7 schools and 5 office blocks from those completed during a certain period-

Schools	28	31	26	27	23	38	37
Office blocks	37	42	34	37	35		

At $\alpha = 0.02$, do the data support the hypothesis that the cost per square meter for the office blocks was greater than that for the schools? [$t_{10}(0.02) = -2.36$] [6]

- (c) The results of a random sample of children with pain from musculoskeletal injuries treated with acetaminophen, ibuprofen, or codeine are shown in the table.

	Acetaminophen	Ibuprofen	Codeine
Significant Improvement	78	61	86
Slight Improvement	58	84	62
No Improvement	67	54	76

At $\alpha = 0.10$, is there enough evidence to conclude that the treatment and response are independent? [$\chi^2_4(0.10) = 7.78$] [6]

- Q.5 (a) Show that $Y(t) = X(t)\cos(\omega_0 t + \theta)$ is a WSS process, where $X(t)$ is a WSS process, θ is uniformly distributed random variable in $(-\pi, \pi)$ and ω_0 is a constant. [6]

- (b) On January 1 this year bakery A had 40% of its local market share while the other two bakeries B and C had 40% and 20 % of the market share respectively. Based upon a study by a marketing research firm, the following facts were compiled- Bakery A retains 90% of its customers while gaining 5% of B's customers and 10% of C's customers. Bakery B retains 85% of its customers while gaining 5% of A's customers and 7% of C's customers. Bakery C retains 83% of its customers and gains 5% of A's customers and 10% of B's customers. What will each firms share be on January 1 next year and what will each firm's market share be at equilibrium? [6]

- (c) Define (i) Ergodic (irreducible) Markov chain (ii) Regular Markov chain. Show that the Markov chain given by the transition probability matrix P is ergodic but not regular. [6]

$$P = \begin{bmatrix} 0 & 0.15 & 0.85 & 0 \\ 0.32 & 0 & 0 & 0.68 \\ 0.27 & 0 & 0 & 0.73 \\ 0 & 0.64 & 0.36 & 0 \end{bmatrix}$$
